

# Referral Program Impact Analysis

## Company XYZ

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## **Executive Summary:**

Company XYZ launched a referral program on October 31, 2015, offering users a \$10 credit for successful referrals resulting in purchases. Following the launch, the company observed increases in users, transactions, and revenue.

This analysis evaluated platform performance before and after the referral program launch using transaction-level data and statistical analysis techniques.

Key findings included:

- User growth increased significantly after launch
- Revenue increased by approximately 16.8%
- Referral-generated revenue exceeded estimated referral incentive costs
- Statistical testing did not find sufficient evidence to conclusively attribute revenue growth solely to the referral initiative

Overall, the referral program demonstrated positive directional performance, though further controlled experimentation is recommended.

## **Business Problem:**

Company XYZ introduced a referral program designed to accelerate user growth and increase purchases through customer referrals. Under the program, users received a \$10 credit when a referred user completed a purchase.

Following the launch, management sought to evaluate whether the referral initiative positively impacted platform performance and justified the associated incentive costs.

## **Objectives:**

The objective of this analysis was to evaluate the impact of the referral program on:

- User growth
- Transaction volume
- Revenue performance
- Referral participation
- Estimated return on investment (ROI)

## Methodology:

The analysis compared platform performance before and after the referral program launch date of October 31, 2015.

The dataset was divided into two groups:

- Before Launch
- After Launch

Key performance indicators were aggregated and compared across both periods. Statistical hypothesis testing was performed using a two-sample t-test on daily revenue values to evaluate whether observed changes were statistically significant.

## Data Preparation:

Data preparation steps included:

- Importing the dataset into Python using pandas
- Converting date fields into datetime format
- Splitting observations into before and after launch periods
- Aggregating daily metrics
- Identifying referral transactions
- Calculating referral incentive costs

## Key Findings:

| Metric       | Before Launch | After Launch | Change  |
|--------------|---------------|--------------|---------|
| Unique Users | 5,000         | 18,397       | +267.9% |
| Transactions | 47,341        | 50,000       | +5.6%   |
| Revenue      | \$2,006,396   | \$2,344,003  | +16.8%  |

## Referral Program Performance:

| Metric                   | Value       |
|--------------------------|-------------|
| Referral Transactions    | 28,017      |
| Referral Revenue         | \$1,315,787 |
| Estimated Referral Costs | \$280,170   |

## Return on Investment:

| Metric                  | Calculation                                | Value       |
|-------------------------|--------------------------------------------|-------------|
| Referral Revenue        | Sum of referral transaction revenue        | \$1,315,787 |
| Referral Transactions   | Total successful referral purchases        | 28,017      |
| Referral Incentive Cost | $28,017 \times \$10$                       | \$280,170   |
| Net Referral Revenue    | $\$1,315,787 - \$280,170$                  | \$1,035,617 |
| Estimated ROI           | $(\$1,315,787 - \$280,170) \div \$280,170$ | 369.6%      |

The referral program generated approximately \$1.32 million in referral-associated revenue while incurring an estimated \$280 thousand in referral incentive costs. Based on these calculations, the program produced an estimated directional ROI of approximately 370%.

This suggests that for every \$1 spent on referral incentives, the company generated approximately \$3.70 in referral-associated revenue. Although the results appear financially promising, the absence of a randomized control group limits the ability to conclusively attribute all observed revenue growth directly to the referral initiative.

## Statistical Analysis:

| Metric      | Value   |
|-------------|---------|
| T-Statistic | -1.1138 |
| P-Value     | 0.2703  |

The statistical analysis did not produce a p-value below the standard significance threshold of 0.05. Therefore, the analysis did not find sufficient statistical evidence to conclude that the referral program directly caused the observed increase in revenue.

This could be due to the size of the dataset, not that there is not a standard significance.

## **Limitations:**

Because the referral program was launched globally without a randomized control group, the analysis cannot fully isolate the causal effect of the referral initiative from external factors such as seasonality, marketing campaigns, or organic growth trends.

## **Recommendations:**

Recommendations include:

- Continue monitoring referral program performance
- Analyze long-term retention of referred users
- Optimize referral incentives and onboarding flows
- Conduct a randomized A/B test for stronger causal analysis

## **Conclusion:**

The referral program demonstrated positive directional business impact through increases in users, transactions, and revenue. Although statistical testing did not provide definitive causal evidence, the program appears promising and warrants continued optimization and controlled experimentation.